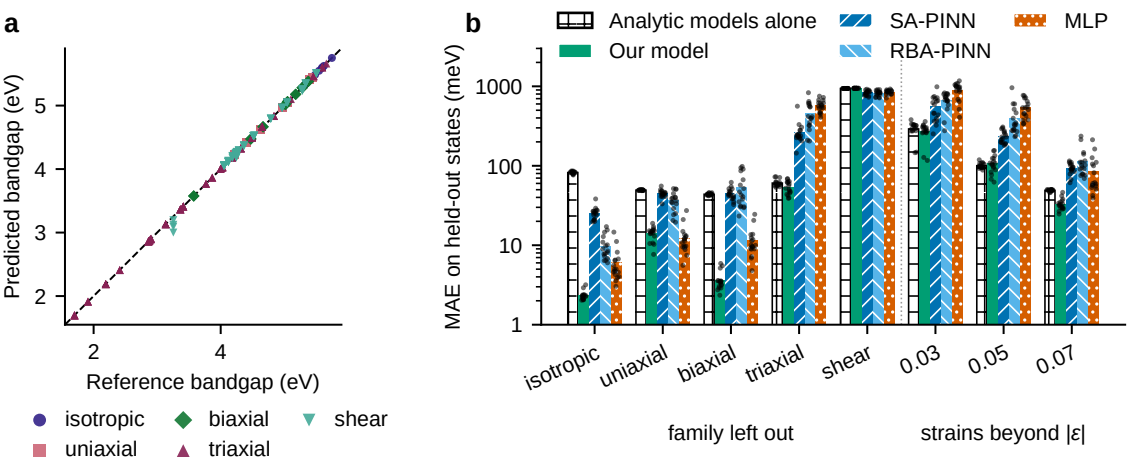

a
Predicted bandgap (eV)
Reference bandgap (eV)
2
3
4
5
2
4
isotropic
uniaxial
biaxial
triaxial
shear
b
Analytic models alone
Our model
SA-PINN
RBA-PINN
MLP
MAE on held-out states (meV)
1000
100
10
1
isotropic
uniaxial
biaxial
triaxial
shear
0.03
0.05
0.07
family left out
strains beyond |ε|